

# Project Proposal

## Visual Augmentation of Audio Semantics for Accessibility

Adam Gavriely

August 6, 2026

## 1 Project Overview

### 1.1 Motivation

Accessibility solutions for deaf and hard-of-hearing users primarily focus on spoken language through subtitles or sign language interpretation. While these approaches effectively communicate dialogue, they often fail to convey the rich semantic information contained in ambient and environmental sounds.

Examples include flowing water, approaching footsteps, distant sirens, birds singing, rain, crowd reactions, machinery, or other background sounds that contribute to the overall understanding and atmosphere of a scene. Although such sounds may not always be essential for understanding the dialogue, they often provide important contextual information that enhances the viewing experience.

This project investigates whether these missing audio cues can be complemented through automatically generated visual content, thereby enriching the accessibility experience beyond traditional subtitles or sign language.

The proposed system aims to communicate such information through complementary visual augmentations, providing deaf and hard-of-hearing users with a viewing experience that more closely resembles that of hearing viewers.

### 1.2 Problem Statement

The objective of this project is to design and implement a system that improves accessibility for deaf and hard-of-hearing users by augmenting an existing video with complementary visual information derived from its audio stream.

Rather than replacing the original video or generating an entirely new visual scene, the proposed system aims to identify semantically relevant information contained in the audio, with particular emphasis on ambient and environmental sounds that may not be fully conveyed through the visual modality. The identified information is then represented through automatically generated visual augmentations displayed alongside the original video.

The proposed approach investigates whether such visual augmentations can enrich scene understanding beyond traditional accessibility solutions such as subtitles while complementing, rather than replacing, sign language interpretation.

### 1.3 Project Scope

The project focuses on investigating the feasibility of semantic audio augmentation using existing foundation models rather than developing new machine learning architectures.

The expected contributions include:

- Formulating the semantic audio augmentation task for accessibility.

- Designing an inference pipeline based on existing open-source foundation models.
- Constructing an evaluation benchmark.
- Developing an automatic evaluation protocol.

## 2 Research Questions

The project seeks to investigate the following research questions:

- Which types of audio information contribute most to scene understanding when presented visually?
- What level of semantic granularity should be used when representing ambient and environmental sounds?
- Can automatically generated visual augmentations improve accessibility beyond traditional subtitles?
- How should semantic audio augmentation systems be evaluated?
- How does the proposed pipeline compare with representative audio-to-visual generation approaches?

## 3 Input and Output Definition

### 3.1 Input

The proposed system receives a short video clip consisting of synchronized visual and audio streams.

**Video** The visual stream provides the scene context and is used to determine which information is already visible to the viewer.

Candidate video sources include:

- Movie and television scenes
- Documentaries
- Real-world recordings

**Audio** The accompanying audio stream may contain one or more of the following components:

- Spoken language
- Ambient sounds
- Environmental sounds
- Acoustic events
- Background music

### 3.2 Output Definition

The output of the proposed system is a set of complementary visual augmentations that are displayed alongside the original video.

The objective is not to replace the original visual content, but to supplement it with semantic information contained in the audio that may not be apparent from the video alone.

Initially, the project focuses on generating static storyboard-style images. Depending on project progress, the representation may later be extended to short generated video segments.

The generated visual content may represent ambient sounds, environmental events, invisible sound sources, or additional contextual information inferred from the audio.

## 4 Methodology

The exact architecture and model selection will be determined following the literature review and in consultation with the supervisor.

The proposed methodology follows a modular inference pipeline in which each component can be independently evaluated or replaced.

Candidate models listed below represent possible implementations and may change during the project.

### 1. Audio Extraction

Extract the audio track from the input video and convert it into a standardized format suitable for downstream processing.

Candidate tools:

- FFmpeg

### 2. Video Understanding

Extract high-level scene context from the visual stream to support cross-modal reasoning. The visual information provides contextual cues that help interpret the semantic meaning of the accompanying audio.

Candidate models:

- Qwen2.5-VL
- LLaVA
- InternVL

### 3. Speech Recognition

Transcribe spoken language contained in the audio stream.

Candidate models:

- Whisper Large V3
- Qwen2-Audio

### 4. Audio Event Detection

Detect and classify non-speech audio events and ambient sounds.

Candidate models:

- BEATs

- PANNs
- AudioSet-based models

## 5. Cross-Modal Semantic Analysis

Combine the extracted scene context, spoken language, and detected audio events into a unified semantic representation.

The objective of this stage is to determine which audio cues are semantically meaningful for accessibility and how they should be represented visually. Particular emphasis is placed on ambient and environmental sounds that enrich scene understanding beyond spoken dialogue.

Candidate models:

- Qwen3
- Llama 3.1

## 6. Visual Augmentation Generation

Generate complementary visual illustrations representing the selected semantic information.

The generated content is intended to supplement the original video rather than replace it.

Candidate models:

- FLUX.1
- Stable Diffusion XL

## 7. Evaluation

Evaluate whether the generated visual augmentations successfully communicate the intended semantic information.

The evaluation methodology will be refined during the project and is expected to include automatic model-based evaluation together with qualitative human assessment.

# 5 Benchmark Dataset

The benchmark will consist of approximately **300 short video clips**, each with a duration of approximately 10–20 seconds.

The objective of the benchmark is to evaluate the proposed system under realistic conditions in which the audio stream contains semantic information that is only partially represented by the visual modality.

## 5.1 Evaluation Scenarios

The benchmark will evaluate the proposed system under a diverse set of realistic acoustic scenarios.

### • Ambient Environmental Sounds

Scenes containing environmental sounds that contribute to the atmosphere or context of the scene (e.g., flowing water, rain, wind, birds, traffic, crowd ambience).

### • Acoustic Events

Scenes containing localized sound events such as alarms, sirens, explosions, glass breaking, applause, laughter, or crying.

- **Mixed Audio Scenes**

Scenes containing a combination of spoken dialogue, ambient sounds, and acoustic events that jointly contribute to scene understanding.

## 5.2 Benchmark Construction

The benchmark will be manually curated from publicly available datasets.

Candidate clips will be selected according to the following criteria:

- Rich semantic audio content.
- Audio information that is only partially represented by the visual stream.
- High-quality synchronized audio and video.
- Diversity of environments and acoustic conditions.

## 5.3 Candidate Data Sources

Candidate benchmark sources include:

- **MovieNet**  
Movie scenes containing dialogue, ambient sounds, and complex acoustic environments.  
<https://movienet.github.io/>
- **VGGSound**  
Real-world videos containing diverse environmental sounds and synchronized audio.  
<https://www.robots.ox.ac.uk/~vgg/data/vggsound/>
- **UnAV-100**  
Large-scale audio-visual event dataset containing realistic scenes with multiple concurrent sound events.  
<https://unav100.github.io/>

The final benchmark composition will be determined following the literature review.

# 6 Evaluation

The objective of the evaluation is to determine whether the generated visual augmentations successfully communicate semantic information contained in the audio that is not sufficiently conveyed by the original video.

## 6.1 Automatic Evaluation

The proposed evaluation consists of the following steps:

1. Analyze the original video and audio.
2. Generate complementary visual augmentations.
3. Use a Vision-Language Model (VLM) to describe the augmented scene.
4. Compare the generated description with a semantic reference obtained from the original multimodal input using a Large Language Model.
5. Evaluate the semantic consistency using an independent LLM-based judge.

The exact evaluation protocol will be refined during the project.

## 6.2 Human Evaluation

A qualitative user study may be conducted to evaluate the usefulness of the proposed visual augmentations.

Participants will compare different accessibility settings, including subtitles alone and subtitles combined with the generated visual augmentations, and answer questions regarding scene understanding. The study will assess whether the proposed approach provides additional semantic information beyond conventional accessibility solutions.

## 7 Baseline Systems

The proposed approach will be compared against representative baseline systems to assess the contribution of explicitly reasoning about the relationship between the visual and audio modalities.

Candidate baseline systems include:

- **Direct Audio-to-Image Generation**

Generate a visual representation directly from the audio stream without explicitly analyzing the visual content or identifying missing semantic information.

- **Audio Captioning**

Generate textual descriptions of the audio without producing additional visual augmentations.

- **Proposed Method**

Analyze both the video and audio streams, identify semantic information that is insufficiently conveyed visually, and generate complementary visual augmentations.

Additional baseline systems will be selected following the literature review.

## 8 Expected Contributions

The expected contributions of this project include:

- Formulation of semantic audio augmentation as an accessibility task.
- Design and implementation of an inference pipeline using existing open-source foundation models.
- Construction of a benchmark for evaluating semantic audio augmentation.
- Development of an automatic evaluation methodology.
- Experimental comparison with representative baseline systems.

## 9 Literature Review

The literature review will focus on the following research areas:

- Audio Event Detection
- Audio Captioning
- Audio Understanding

- Vision-Language Models
- Multimodal Large Language Models
- Audio-to-Image Generation
- Audio-to-Video Generation
- Accessibility Technologies

## 10 Related Work

**Audio-to-Visual Generation** Recent work has investigated generating images or videos directly from audio. Representative examples include Sound2Scene and Audio-to-Video generation methods, which aim to synthesize plausible visual content corresponding to an input audio signal.

The proposed work addresses a different objective. Rather than generating a new visual scene, it investigates how semantic information contained in the audio stream can be used to augment an existing video with complementary visual cues that improve accessibility for deaf and hard-of-hearing users.

**Accessibility** Current accessibility solutions primarily rely on subtitles and sign language interpretation.

Subtitles effectively communicate spoken language but generally provide limited information about ambient sounds, environmental events, or the overall acoustic atmosphere. Sign language can communicate richer semantic information than subtitles, yet automatic sign language generation remains an active research challenge.

The proposed work investigates an alternative accessibility approach in which missing audio semantics are complemented through automatically generated visual content.

## 11 Project Deliverables

- Source code
- Curated benchmark dataset
- Automatic evaluation protocol
- Experimental evaluation
- Technical report
- Prototype accessibility augmentation system